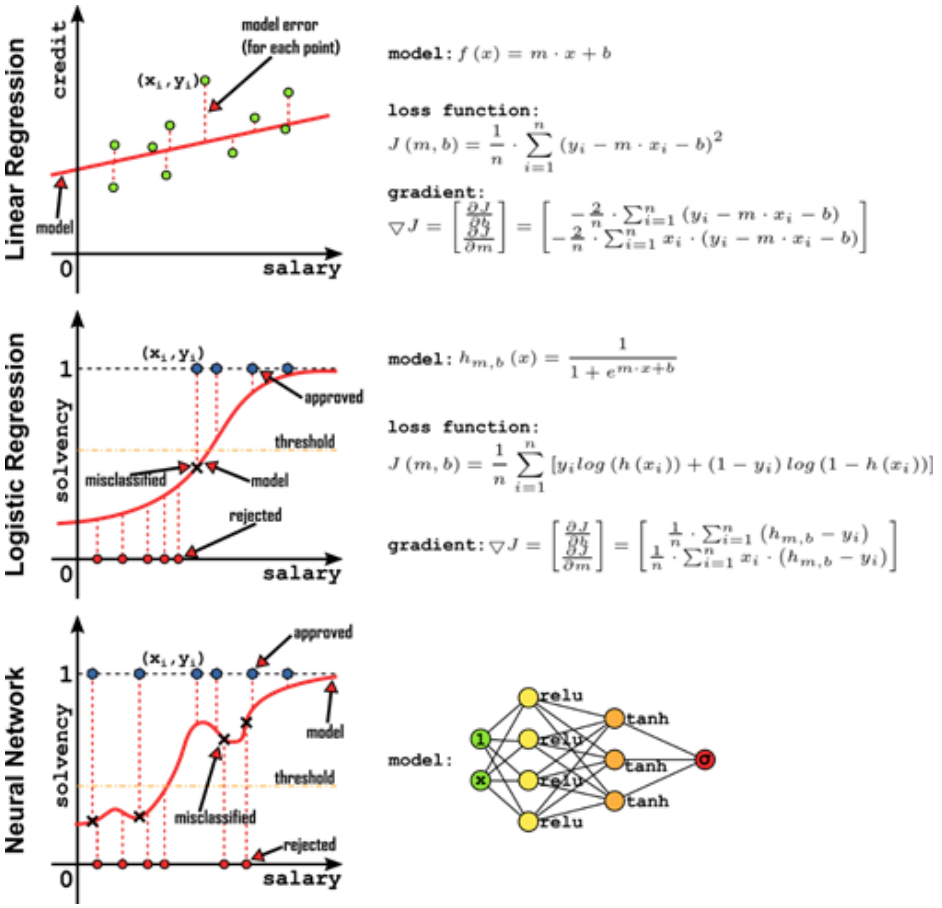


Figure 16.17 A few approaches used in supervised machine learning: all of them use gradient descent for learning. Note that the models are arbitrarily chosen, and also for the loss functions there are other possible choices. Gradients and loss functions for neural networks are omitted for the sake of brevity.
While only supervised learning is shown here, there are also clustering algorithms that leverage gradient descent.

Take, for instance, supervised learning (whose most prominent examples are shown in figure 16.17): this is a field where gradient descent is widely applied!

The goal of supervised learning is to develop a mathematical model[14] that can succinctly describe the dataset, and it's able to predict the output for new, never seen inputs, being it a real